

CSE 115 - PROGRAMMING LANGUAGE I

PROJECT UPDATE

SCIENTIFIC CALCULATOR

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❖ WEEK-01 (June 16 - June 22)

- After creating the repository for the project, I committed my first folder in the repository.
- Then I added necessary files such as `prithu.c` that holds my codes, and a project update pdf for future.
- After finishing with files organizing, I programmed a function called *`power_root`* to operate exponential and root operations. It has 4 types of function : *`square`*, *`square-root`*, *`exponent of any number`*, *`root of any number`*. This function also handles imaginary numbers.

❖ WEEK-02 (June 23 - June 29)

- Added functionality to handle imaginary numbers in the *`power_root`* function.
- Added basic structure for *`matrices`* function.
- Updated the `readme.md` file in the github project repository
- Added the things below in the *`matrices`* function
 - Function to *`validate matrix's dimensions`*
 - Function to take *`input`* for a matrix

- Function to *print* a matrix
- Function for matrix *addition* and *subtraction*
- Function for matrix *multiplication*

❖ WEEK-03 (June 30 - July 6)

- Added a *matrix_determinant* function in the *matrices* program. For this, I used a recursive function called *get_determinant* and another function called *get_submatrix* to get the matrix after removing the first row and n-th column in the matrix so that the determinant of that submatrix can be found recursively. Base cases in the recursive function are for 2x2 and 1x1 matrices.
- Added a *matrix_inverse* function that uses the determinant to evaluate if the inverse of a matrix even exists. If it does, then it uses a cofactor matrix to store the submatrix's determinant with its sign to store. Then strategically transpose the cofactor matrix and divide each element with the determinant.
- Solved some bugs in the *matrix multiplication* function. These bugs were mainly in showing the output of the multiplied matrix.